Local goodness-of-Fit measures for neural decoding

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Abstract

State-space models are widely used in many fields to infer latent states from observations, but there are still some issues and challenges related to them. First, because latent variables are inherently unobservable, their decoding results cannot be directly validated against observed data. Second, it is often challenging to disentangle prediction errors stemming from model misspecification. Third, such errors can accumulate and propagate over time, amplifying uncertainty in long-term predictions. Moreover, many state-space models are purposefully simplified, accepting some degree of inaccuracy in exchange for greater computational efficiency. To address this challenge, we developed three classes of metrics designed to identify localized periods where model misspecification leads to inaccurate estimates. Each metric assesses the consistency between the information contained in individual observations and the information extracted by the state-space model from a local window of data. These metrics fall into three categories: 1) a set of divergence measures, specifically F-divergences, which compare the distributions of the latent state process directly; 2) measures of overlap between estimated credible regions obtained from the state distributions; and 3) predictive checks for individual observations, based on samples from the estimated state process. We applied these metrics to the problem of estimating nonlocal spatial representations in both simulated and real hippocampal spiking data, demonstrating the strengths and limitations of our approach.

1 Introduction

Many brain states, such as those related to memory, emotion, and attention, are not directly observable. They are internal states that are not solely determined by the external world and can change over time. This makes them challenging to understand. However, these brain states are critical for understanding how the brain supports complex behavior, because they go beyond simple stimulus and response relationships. A major challenge for systems neuroscience is to relate neural activity (spikes, spike waveforms, calcium traces, etc.) to these latent dynamic brain states.

State space models are a powerful way to understand how neural activity relates to latent dynamic brain states. They have been applied to diverse neural systems such as gustatory activity (Jones et al., 2007), hippocampal replay dynamics (E. L. Denovellis et al., 2021), and hypothalamic activity dynamics (Vinograd et al., 2024). The core assumption of these state space models is that complex, high dimensional neural activity can be related to low dimensional, latent states. State space models aggregate information over time about the latent states, weighing both the information from the data at the current time (normalized likelihood distribution) with the accumulated information used to predict the latent state (prediction distribution). This powerful combination enables robust estimation of the latent state, effectively balancing current information with the accumulated history of the system and prior knowledge.

Despite the wide use of state space models in neuroscience, methods to validate state space models remain underdeveloped and underutilized. One reason is that latent states are inherently difficult to evaluate (Newman et al., 2014). Without ground truth, model validity can only be inferred indirectly—from relative likelihoods under multiple models, from the quality of reconstructed signals, or from performance on downstream tasks. Another challenge is that neural data are dynamic, so model fit may vary substantially across time. Compounding this problem is that common methods of validating models, such as cross-validation and "shuffle" permutation tests, are holistic evaluations of the model – they tell neuroscientists whether or not the model generalizes well or is different from a simplified model with less utility – but they fail to indicate when the model fits poorly or which specific components are responsible. Consequently, these methods can reject models that are intentionally simplified yet scientifically meaningful and offer little guidance for targeted improvement of the model. Many state-space models in neuroscience are intentionally simplified—not because the brain is simple, but because tractable computation and clear interpretation often matter most. For example, some state-space models smooth emotional movement into a low-dimensional trajectory (Deng et al., 2015); others compress field-linked structure into a stationary, locally anchored map (Hsin et al., 2022). Some frameworks simplify both at once, jointly compressing affective trajectories and local field potentials into compact latent states for computation and inference (Tao et al., 2018; Zeng & Eden, 2025). Under standard goodness-of-fit criteria, such simplifications would likely fail.

State space models already contain rich diagnostic information in the form of distributions used in their estimation. One potential solution for more targeted local diagnostic information of state space models would be to utilize the information in the normalized likelihood, which reflects information from only the current neural observations, and prediction distribution, which aggregates information from all past observations based on the state space model structure. When these two distributions agree, the model's prior expectations and data-driven evidence are consistent. When they diverge, the mismatch can reveal where and when and potentially how the model fails to capture the structure of the data. Examining these discrepancies could therefore provide a direct, interpretable measure of model–data agreement—analogous to a local, time-resolved goodness-of-fit test.

To address these gaps, we develop and validate a suite of three complementary goodness-of-fit methods – KL divergence, highest posterior density (HPD) overlap region, predictive checks – designed to provide actionable feedback to neuroscientists. Specifically, each of these methods utilize the mismatch between prediction and likelihood to: (1) identify time periods of poor model–data agreement; (2) distinguish errors arising from the state versus observation components; and (3) provide intuitive visualizations accessible to a broad audience. We demonstrate the utility of these techniques on simulated and real hippocampal data, offering a path for neuroscientists to move from binary model rejection toward an iterative, diagnostic approach to building interpretable theories of neural dynamics.

2 Methods

2.1 State-Space Models

The state-space paradigm is particularly well suited for neural data analysis since many neuroscience experiments focus on internal brain processes that are not directly observable. For example, hippocampal replay of spatial trajectories has been hypothesized to be related to mental exploration of space, but the specific spatial trajectories being replayed are not directly observable by experimentalists (Tao et al., 2018). In such cases, state-space models are used to estimate simultaneously the unobserved internal representations and their relation to neural activity patterns.

For a discrete set of time points, $t_0, \ldots, t_K$, we define an unobserved state process x_k, which influences a neural observation process y_k at each time t_k. We define a state space model by combining a state transition distribution $p(x_k|x_{k-1})$, which defines how the state evolves in time, with an observation likelihood $p(y_k|x_k)$, which defines the neural representation of each state. To simplify the problem, we assume that the state transition depends only on the most recent past value of the state and that the observation likelihood depends only on the current state value. Figure 1 (a) shows a schematic of this state-space model structure.

This state-space model structure allows for iterative estimation of the state process at each time step based on all the neural data up to that time, as shown in Fig. 1 (b). If the distribution of the state at time t_{k-1} has been computed, we can convolve it with the state transition distribution

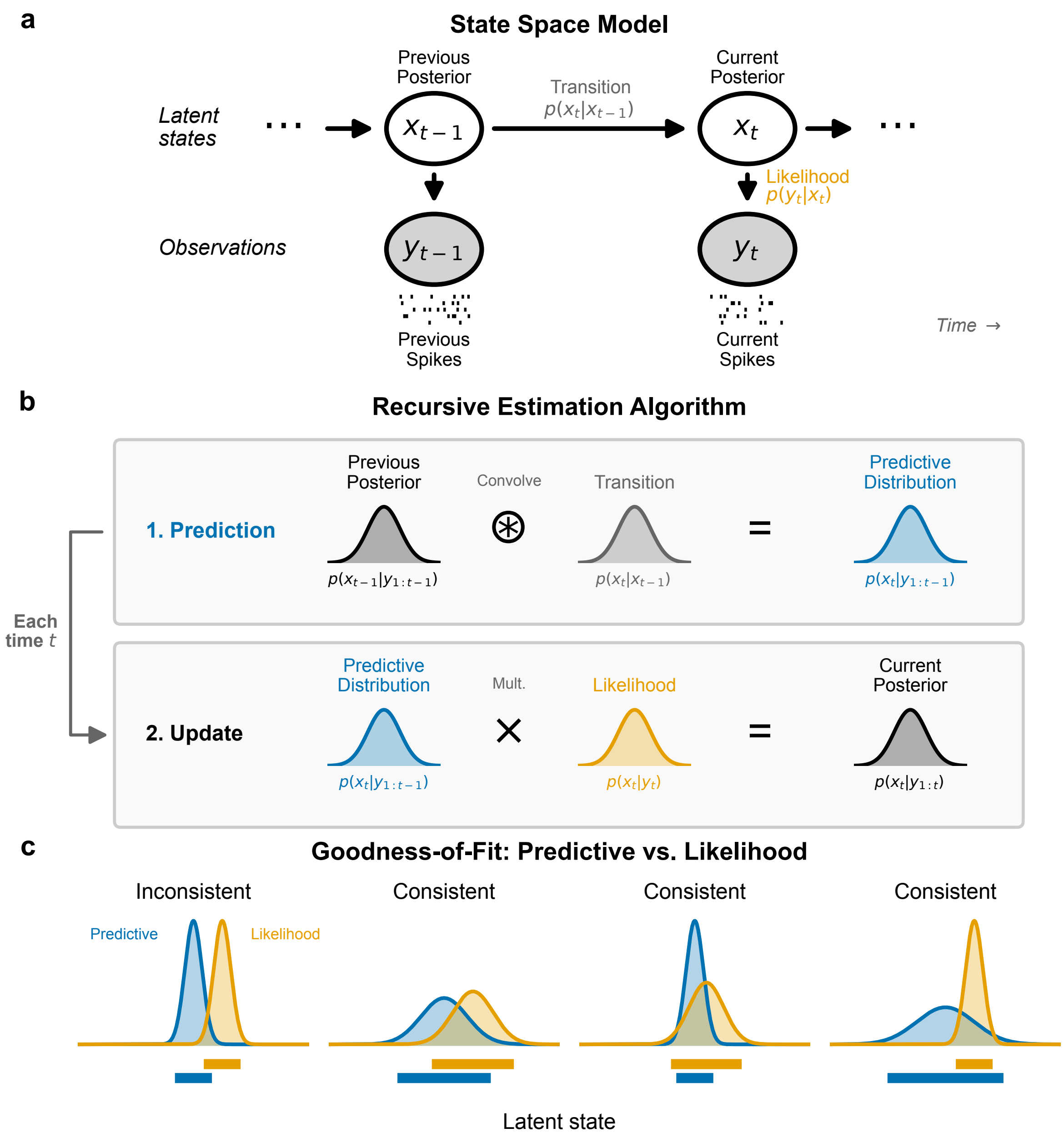

Figure 1: Schematics for state-space model and goodness-of-fit measures. (a) Graphical model depiction of state-space model includes state transition model, $p(x_t|x_{t-1})$, and observation likelihood $p(y_t|x_t)$. (b) These model components are sufficient to compute an iterative expression for the posterior filter distribution at time t_k, $p(x_k|y_{1:k})$ from the distribution at the previous time step. This involves first convolving the previous posterior with the state transition distribution to compute the one-step predictive distribution, $p(x_k|y_{1:k-1})$, and then updating the predictive distribution with information from the most recent spiking by multiplying by the observation likelihood. (c) For each new spike observation, we compute the local goodness-of-fit by assessing the consistency of information between the predictive distribution and likelihood. These are considered inconsistent when they assign high probability to mostly disjoint subsets of the state-space. When the regions with high probability overlap, the distributions are considered consistent, even if the distributions differ in their central tendency or certainty.

to obtain a one-step prediction distribution, $p(x_k|y_{1:k-1}) = \int_{x_{k-1}} p(x_k|x_{k-1}) \cdot p(x_{k-1}|y_{1:k-1}) dx_{k-1}$. This predictive distribution combines the information from all previous observations to compute the distribution of the state at the next time step. We update this predictive distribution with the information from the most recent observation by multiplying by the observation likelihood to compute the posterior distribution of the state given all of the data up to the current timestep, $p(x_k|y_{1:k}) \propto p(y_k|x_k) \cdot p(x_k|y_{1:k-1})$.

Common approaches for assessing the goodness-of-fit of state-space models include examining the cross-validated error measures in the state process when the state is observable (Stone, 1974), cross-validated error measures in the observation process such as the mean-squared prediction error (Hastie, 2009), and the distribution of prediction residuals (Shen & Zhuang, 2024). For many neural modeling problems, state-space models are intentionally simplified and these standard goodness-of-fit measures trivially show lack of fit.

Here we explore a set of local goodness-of-fit measures, each of which compares the prediction distribution to the normalized likelihood of the most recent observation as a function of the state or to that observation itself. Thus, these measures consider the consistency between information accumulated through the state-space model over past spiking with the information from the most recent spiking activity. We do not expect these distributions to align perfectly since they likely contain different amounts of information about the state process. Instead, we assess whether the information these distributions provide about the state and observation processes is consistent in the sense illustrated in Fig. 1 (c). Specifically, we say that the two distributions are inconsistent when their probability mass is concentrated in mostly non-overlapping regions of the state space. This suggests disagreement between the information from past spiking and the information from the most recent spiking. Distributions that largely align, or that are shifted from each other but are supported over largely overlapping regions of state space are considered consistent. Additionally, if one distribution covers a broad region of the state-space and the other covers a much narrower subset of that region, the distributions are considered to be consistent. We expect this situation to arise often since the prediction distribution contains information accumulated from many past spikes.

2.2 Local Goodness-Of-Fit Measures

In this section, we introduce three classes of metrics designed to characterize the local goodness-of-fit of the state-space model.

2.2.1 KL Divergence

One of the most well-known measures of discrepancy between probability distributions is the Kullback-Leibler (KL) divergence (Solomon et al., 1951), defined in this case as the expected excess self-information (log of the density function) when using the likelihood to predict the state in place of the predictive distribution,

$$D_{\mathrm{KL}}(P\|Q) = \int P(x_k)\log\left(\frac{P(x_k)}{Q(x_k)}\right)dx_k, \tag{1}$$

where $P(x_k) = p(x_k|y_{1:k-1})$ is the predictive distribution and $Q(x_k) = \frac{p(y_k|x_k)}{\int p(y_k|x_k)dx_k}$ is the normalized likelihood in x_k given the observed spiking y_k.

Do we use $p(y_k|x_k)$ with a uniform prior instead normalized likelihood?

Figure 2 (a) illustrates the computation of the KL divergence. The top panel shows the predictive distribution and normalized observation likelihood as functions of x_k. The middle

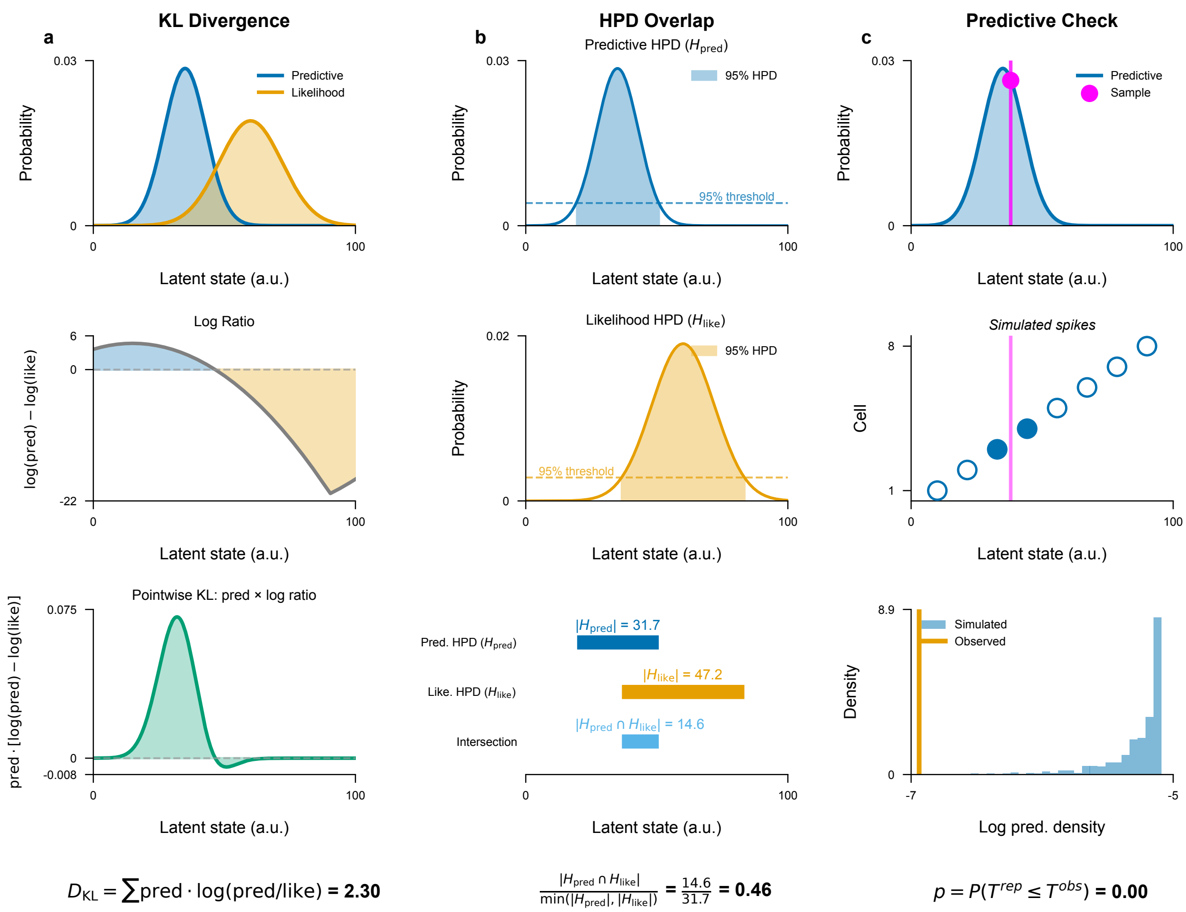

Figure 2: Visualizations of the computation of each local goodness-of-fit metric. (a) The KL divergence compares the predictive distribution (blue) and the observation likelihood (orange) (top panel) by computing the difference in their log density (middle panel) and computing its expectation with respect to the predictive distribution (bottom panel). (b) The HPD overlap is defined by computing the HPD for the predictive distribution (top) and for the observation likelihood (middle) and then computing the size of their intersection relative to the size of the smaller of these two HPD regions (bottom). (c) Predictive checks are computed by sampling values of the state from the predictive distribution multiple times (top), then sampling from the conditional distribution of neurons that could have generated the observed spike given each sampled state value (middle), and then computing the probability that the log of the predictive distribution of the observed spike is greater than the log of empirical predictive distribution of the sampled spikes (bottom).

panel shows the difference in self-information between these distributions, $\log\left(\frac{P(x_k)}{Q(x_k)}\right)$. The bottom panel is this difference in self information weighted by $P(x_k)$, which when integrated gives the value of the KL divergence.

The KL divergence is an asymmetric measure, meaning that $D_{KL}(P||Q) \neq D_{KL}(Q||P)$. We could also compute the KL divergence from the likelihood to the predictive distribution $D_{KL}(Q||P)$ or the mean between these two divergences $\frac{1}{2}(D_{KL}(P||Q) + D_{KL}(Q||P))$. More broadly, the KL divergence is part of a larger family of divergence metrics between distributions, known as F-divergences. Other members include the total variation distance (Bhattacharyya et al., 2023) and the Pearson χ^2 divergence (Crack, 2018). It is simple to extend the KL divergence metric between the predictive distribution and the observation likelihood to any of these other divergence metrics to identify differences in additional features of the distributions.

2.2.2 Overlap Between HPD Regions

Another way to assess the consistency between the predictive distribution and the observation likelihood is to examine the extent to which their highest posterior density (HPD) regions overlap. A substantial intersection suggests that the two distributions convey consistent information, reinforcing the reliability of the inferred latent state. In contrast, minimal or no overlap signals a significant discrepancy, hinting at potential inconsistencies in the local fit of the model. This approach provides an interpretable and intuitive metric of the degree of consistency between these distributions.

We define the overlap between the HPD for the predictive distribution and the likelihood as

$$\text{HPD Overlap}\,(P, Q) = \frac{|\cap\,(HPD_P, HPD_Q)|}{\min\{|HPD_P|\,, |HPD_Q|\}} \tag{2}$$

where HPD_P and HPD_Q are the highest posterior density regions of the predictive density and the normalized observation likelihood respectively and $| * |$ is the volume of the region. The numerator is the size of the overlap between these regions and the denominator is the size of the smaller of the two HPD regions. This normalization ensures that the resulting value is in the range $[0, 1]$, with 0 indicating no overlap, and 1 indicating that one of the HPDs is entirely contained within the other. This means that if, for example, the likelihood produces an HPD that is wider and therefore less informative than the predictive density but that completely contains the HPD of the predictive density then the HPD ratio will be 1, suggesting that the two distributions are statistically consistent. Conversely, if the ratio is close to 0, it indicates little or no overlap, suggesting significant differences between range of values for the state that would be inferred from these two distributions.

Figure 2 (b) illustrates the computation of the HPD overlap. The HPD for the predictive distribution (top) and the normalized likelihood (middle) are computed, and the size of their intersection is normalized by the size of the smaller of the two HPDs (bottom).

2.2.3 Predictive Checks

The previous metrics compared two distributions of the state: one based on the accumulated past information and the other based on the isolated information from the likelihood of the most recent spiking observation. Another approach is predictive checking, which compares the observation itself to a predictive distribution of that observation estimated from the model. Bayarri et al. (2007) proposed Bayesian checking of the second levels of hierarchical models with partial posterior predictive checking. Gemma et al. (2019) used population predictive checking to evaluate model's goodness-of-fit. Guttman (1967) used future observations to evaluate model's goodness-of-fit. Predictive checks have the advantage that they allow for the computation of p-values for individual observations, which are easy to interpret (X. Meng, 1994; A. Gelman, X. Meng and H. Stern, 1996).

Predictive checks for spiking data can be challenging to implement, since in sufficiently small intervals, any spikes are unlikely compared to no spiking at all. One approach to deal with this might be to consider patterns of spikes over longer fixed intervals over which multiple spikes would be expected. Here, we take a simpler approach of focusing only on spike times and measuring the likelihood that an observed spike comes from a particular neuron (for spike-sorted models) or has a particular set of waveform features (for clusterless models).

In particular, let $y_{k,j}$ be the identity of the neuron (spike-sorted model) or a spike waveform feature (clusterless model) for the j^{th} spike occurring in the k^{th} time bin. We define the predictive distribution of the observation based on the state model as

$$\mathrm{f}_{\text{predictive}}\left(y_{k,j}\right) = \int p\left(y_{k,j} \mid x_k\right) p\left(x_k | y_{1:k-1}\right) dx_k \tag{3}$$

to be the predictive distribution of $y_{k,j}$, where $p\left(y_{k,j} \mid x_k\right)$ is the observation process and $p\left(x_k | y_{1:k-1}\right)$ is the one-step prediction distribution of the current state given all previous spikes. We can sample new observations, $\widetilde{y}_{k,j}$ from this predictive distribution, and compute the probability that the observed spike is as likely as the sampled spike.

$$D_{k,j} = Pr\left(\log\left(\mathrm{f}_{\text{predictive}}\left(y_{k,j}\right)\right) \geq \log\left(\mathrm{f}_{\text{predictive}}\left(\widetilde{y}_{k,j}\right)\right)\right). \tag{4}$$

The log terms in Equation (4) are unnecessary, but suggest the method to estimate $D_{k,j}$. We sample x_k multiple times from the one-step prediction distribution, $p\left(x_k | y_{1:k-1}\right)$ and use these to generate samples of the spike observations, $\widetilde{y}_{k,j}$ from the predictive distribution. We then compute the fraction of these samples whose log likelihood is smaller than or equal to that of the observed spike.

Figure 2 (c) illustrates the computation of a predictive p-value for spike sorted data. Many values of the state are sampled from the predictive distribution (top), each of which is used to sample a neuron that could have generated the observed spike (middle). The log likelihood of the observed spike is then compared to an empirical cumulative distribution of the log likelihoods of the sampled spikes (bottom) leading to a goodness-of-fit metric that can be interpreted as a p-value for that individual spike. High p-values suggest that the observed spike is consistent with the predictive distribution of spikes and a p-value near zero suggests that the observed spike is unexpected based on the predictive distribution computed from the past spiking and the model.

3 Simulation Study

To demonstrate the performance of our local goodness-of-fit metrics in real-world applications, we designed a simulation study, generated neural signal data, constructed state-space model, and applied our metrics to evaluate the model's local goodness-of-fit. We established a framework to analyze spatial mapping in the hippocampus of a rat. Specifically, we simulated data that approximates the rat's movement and the corresponding hippocampal spike activity. Our simulation will encompasses both local and non-local states, which, while resembling real-world rat behavior and neural signals to some extent, may also deviate from them. Our goal is to decode the latent variables using an intentionally simplified model applied to data generated by a more complex, true generative process. We then utilize local goodness-of-fit metrics to assess the model's alignment with observed data, identifying periods of potential misspecification and evaluating the model's reliability over time.

3.1 Generative Model

We assume that the data are observed over a set of discrete time points $t_0, t_1, \ldots, t_k, \ldots, t_K$. We define a discrete variable process s_k, called the locality state. This state defines a set of dynamic models for the spiking data at each time step t_k. We label the values that s_k can take as follows.

$$s_k = \begin{cases} 0, & \text{local state ;} \\ 1, & \text{non-local state with small variance ;} \\ 2, & \text{local state with some non-spatial hippocampal spikes ;} \\ 3, & \text{non-local state with drift diffusion, large variance and low hippocampal spike rate ;} \\ 4, & \text{non-local state with drift diffusion and high hippocampal spike rate;} \end{cases} \tag{5}$$

This is a latent state process, so we cannot directly observe the value of s_k. Instead, we will define another set of latent processes that link this state to the observation processes. We assume the probability of being in any state at time t_k only depends on the state at the previous time t_{k-1}. To simplify the situation, except state 0, we also assume that all other states can only originate from or transition to state 0.

We link the locality state to the hippocampal spiking activity from the population of place cells by defining a continuous state variable x_k, which captures the factors that influence spiking activity. When the rat is in a local state, the neurons have place fields which fire as a function of the linearized position at time t_k. We let o_k be the observed linearized position at time t_k. Therefore, whenever $s_k = 0$, we set $x_k = o_k$. When $s_k \neq 0$ or 2, we treat x_k as an unobserved, latent variable.

When the locality state s_k changes from a local state into a non-local one, we allow x_k to correspond to any location, not just the rat's current location. The distribution of possible initial locations when the local state switches to a non-local state is assumed to be uniform over the entire environment. We assume that as long as the rat stays in the non-local state, its trajectory is assumed to be a random walk, whose variance and drift parameters are determined by the value of s_k. Mathematically, the dynamics of x_k are defined by the following state transition model.

$$p\left(x_k \mid x_{k-1}, s_k, s_{k-1}\right) = \begin{cases} \delta\left(o_k\right), & \text{if } s_k = 0 \text{ or } s_k = 2; \\ \mathcal{U}(a,b), & \text{if } s_k = 1 \text{ or } s_k = 3 \text{ or } s_k = 4 \\ & \text{and } s_{k-1} = 0; \\ \mathcal{N}\left(x_{k-1}, \sigma\right), & \text{if } s_k = 1 \\ & \text{and } s_{k-1} = 1; \\ \mathcal{N}\left(x_{k-1}, \sigma'\right) + \beta\left(t_k - t_{k-1}\right), & \text{if } s_k = 3 \\ & \text{and } s_{k-1} = 3; \\ \mathcal{N}\left(x_{k-1}, \sigma\right) + \beta\left(t_k - t_{k-1}\right), & \text{if } s_k = 4 \\ & \text{and } s_{k-1} = 4; \end{cases} \tag{6}$$

In (6), when the rat is in a local state, i.e. $s_k = 0$ or 2, we just use the observed linearized position at t_k to describe x_k. When the locality state changes from local into non-local, we allow x_k to correspond to any location, not just the rat's current location. The distribution of possible initial locations for the non-local event is assumed to be uniform over the entire environment. We assume that as long as the rat stays in the non-local state with small variance, the trajectory of x_k is continuous, and use a random walk with smaller variance to model transitions. While the rat stays in non-local state with drift diffusion, large variance and low hippocampal spike rate, this trajectory of x_k is characterized by a random walk exhibiting higher variance and a drift term. At the same time, we would generate relatively fewer hippocampal spikes during this state. While the rat remains in a non-local state with drift diffusion and high hippocampal spike rate, the trajectory of x_k is described as a random walk with a drift component. The variance in the random walk as well as the hippocampal spike intensity is the same as that of non-local state with small variance.

The spiking activity of each tetrode is modeled as a marked Poisson process, which defines the probability distribution of observing a spike with particular waveform features as a function of x_k. When $s_k = 0$ or 2, x_k is the rat's actual position and so this model describes the place field properties of the entire population. When $s_k = 1$ or 3 or 4, x_k corresponds to an internal cognitive representation of position, and this model describes the pattern of reactivation of spiking during non-local state. In general, the marks can be any waveform features, up to the full waveform function. Here, we use the maximum amplitude of the spike on each of the 4 electrodes comprising the tetrode to define a 4-dimensional mark. The data comprises the spike time and the value of this mark for each observed spike. The marked point process model for the neural population recorded by i^{th} tetrode is defined by a joint mark-intensity function $\lambda_i\left(t, \vec{m}\right)$, which expresses the probability of observing a spike with features in a neighborhood of $\vec{m}$ in an interval around t. We construct a joint mark-intensity model by expressing $\lambda_i\left(t, \vec{m}\right)$

as a function of s_k and x_k.

We assumed that the spike times and waveforms for all the tetrodes followed a joint mark intensity that included two modes corresponding to two place cells. The parameters in the joint mark intensity vary depending on the locality state, and as the latent spatial variable shifts, the intensity function adjusts accordingly.

$$\begin{aligned}\lambda_i\left(t_k,\vec{m}\right) = &\left(1-\omega^i_{s_k}\right)\sum_{j=1}^{2}\exp\left\{\alpha^i_{s_k,j}-\frac{1}{2}\left(\begin{array}{c} x_k-\mu^i_{x,j} \\ \\ \vec{m}-\mu^i_{\vec{m},j}\end{array}\right)'\Sigma^{-1}\left(\begin{array}{c} x_k-\mu^i_{x,j} \\ \\ \vec{m}-\mu^i_{\vec{m},j}\end{array}\right)\right\} \\ &+\omega^i_{s_k}\sum_{j=1}^{2}\exp\left\{\alpha^i_{s_k,j}-\frac{1}{2}\left(\vec{m}-\mu^i_{\vec{m},j}\right)'\Sigma_m^{-1}\left(\vec{m}-\mu^i_{\vec{m},j}\right)\right\}\end{aligned} \tag{7}$$

where $i = 1, 2, \cdots, E$ and E is the number of tetrodes. Σ is the covariance diagonal matrix with diagonal elements $(\sigma_x^2, \sigma_m^2, \sigma_m^2, \sigma_m^2, \sigma_m^2)$. We chose this form of Σ with zero off-diagonal elements to simplify the scenarios to allow for analytic solutions. Σ_m is a diagonal matrix with diagonal elements $(\sigma_m^2, \sigma_m^2, \sigma_m^2, \sigma_m^2)$. $\omega^i_{s_k}$ can range from 0 to 1. When $s_k \neq 2$, we set $\omega^i_{s_k} = 0$ for $i = 1, 2, \cdots E$. By doing this, we ensure that the generated hippocampal spikes carry spatially relevant information. Otherwise, we set $\omega^i_{s_k} \neq 0$ to obtain some hippocampal spikes without spatial content. When $s_k \neq 3$, we set $\alpha^i_{s_k,j}$ to be a larger value for $i = 1, 2 \cdots E$ and $j = 1, 2$. Otherwise, $\alpha^i_{s_k,j}$ is set to be a relatively small value to obtain fewer spikes in this period. We express the joint mark intensity as a function of time t_k and waveform features $\vec{m}_k$. In reality, it depends on both the latent locality state s_k and the latent spatial content x_k.

3.2 Estimation Model

To evaluate our local goodness-of-fit approaches, we intentionally decode the simulated data with a simple and misspecified model for the latent variables s_k and x_k. As for the latent locality state s_k, instead of 5 discrete states, this model assumes only two states: the local state and the non-local state with small variance. It is defined as follows:

$$s_k = \begin{cases} 0, & \text{local state ;} \\ 1, & \text{non-local state with small variance ;} \end{cases} \tag{8}$$

Similarily, we simplify the state model for x_k as follows.

$$p\left(x_k \mid x_{k-1}, s_k, s_{k-1}\right) = \begin{cases} \delta\left(o_k\right), & \text{if } s_k = 0; \\ \mathcal{U}(a, b), & \text{if } s_k = 1 \\ & \text{and } s_{k-1} = 0; \\ \mathcal{N}\left(x_{k-1}, \sigma\right), & \text{if } s_k = 1 \\ & \text{and } s_{k-1} = 1 \end{cases} \tag{9}$$

This model simplifies the distribution of x_k, setting x_k equal to o_k in the local state while assuming a simple random walk model during non-local state. In addition to the incorrectly assumed state model, we can also employ this misspecified observation process to decode the observed hippocampal spikes and evaluate the local goodness-of-fit with our established metrics. For the joint intensity function of hippocampal spike data, we continue to assume a strong correlation between waveform features and spatial content. In other words, no matter what the locality state is, we utilize the joint intensity function defined by equations (7) with $\omega^i_{s_k} = 0$ for $i = 1, 2 \cdots, E$. This flawed assumption results in an inaccurate observation process.

If the observed spiking data from i^{th} tetrode in the interval $[t_{k-1}, t_k)$ includes ΔN^i_k spikes with marks $(\vec{m}^i_{k,1}, \vec{m}^i_{k,2}, \cdots \vec{m}^i_{k,\Delta N^i_k})$, and we assume independence between tetrodes, the likelihood of the observed spiking is expressed as:

$$L(x_k) \propto \prod_{i=1}^{E} \prod_{j=1}^{\Delta N^i_k} \lambda_i\left(t_k, \vec{m}^i_{k,j}\right) \exp\left[-\Lambda_i\left(t_k\right) \Delta_k\right] \tag{10}$$

where E is the number of tetrodes, $\Delta_k = t_k - t_{k-1}$, and $\Lambda_i(t_k) = \int_{\mathcal{M}} \lambda_i\left(t_k, \vec{m}_{k,j}^i\right) d\vec{m}_{k,j}^i$ is the ground intensity.

Based on the oversimplified decoding model we have defined, we derive the joint filter probability distribution for all latent variables, given all the observed signals up to the current time i.e. $p(s_k, x_k \mid \Delta N_{0:k}^{(1:E)}, \vec{m}_{0:k})$. This allows us to compute the marginal probability that a non-local event with small variance is occurring at each instant as well as the marginal distributions of the latent variables representing the spatial content of each non-local event.

First, using Bayes's rule, combined with the assumptions in this work (Eden et al., 2018), we obtain an iterative expression for the filter distribution.

$$p\left(s_k, x_k \mid \Delta N_{0:k}^{(1:E)}, \vec{m}_{0:k}\right) \propto p\left(\Delta N_k^{(1:E)}, \vec{m}_k \mid x_k\right) \times \sum_{s_{k-1}} \int_{x_{k-1}} Pr\left(s_k \mid s_{k-1}\right) p\left(x_k \mid x_{k-1}, s_k, s_{k-1}\right) p\left(s_{k-1}, x_{k-1} \mid \Delta N_{0:(k-1)}^{(1:E)}, \vec{m}_{0:(k-1)}\right) dx_{k-1} \tag{11}$$

The last term on the right side of (11) is the filter distribution from the previous time step, which is multiplied by the state transition for each latent state and integrated and summed over the previous latent variables to produce the one-step prediction probability distribution. This distribution is then multiplied by the likelihood of the spiking at the current time step, based on the observation model, to compute the filter distribution at the current time step. We select an initial distribution for the states at time t_0, and iterate through (11) to compute the filter distribution at all time steps. We define a non-local event as an extended period when the filter probability of non-local state $\Pr\left(s_k \mid \Delta N_{0:k}^{(1:E)}, \vec{m}_{0:k}\right)$ exceeds a threshold (eg. 0.99, 0.95, 0.9).

In order to define the metrics, we use the following marginal distributions for x_k: the filter probability distribution $p\left(x_k \mid \Delta N_{0:k}^{(1:E)}, \vec{m}_{0:k}\right)$, the one-step predictor probability distribution $p\left(x_k \mid \Delta N_{0:(k-1)}^{(1:E)}, \vec{m}_{0:(k-1)}\right)$ and the isolated probability distribution $p\left(x_k \mid \Delta N_k^{(1:E)}, \vec{m}_k\right)$ based on the current spiking information.

The first metric we define is the KL divergence between the filter distribution of x_k and the isolated distribution. A larger divergence indicates a substantial discrepancy between the distribution inferred from information up to now using the estimation model and the distribution derived from our current observations, suggesting potential issues with local model fitting. To extend this metric for evaluating each detected non-local event, we define a weighted average of the KL divergence between the filter and isolated distributions as shown in equation (**??**). Only intervals containing at least one spike are included in this evaluation; intervals without any spike are excluded from the evaluation.

The second metric we define is the 95% HPDROR between one-step predictor distribution of x_k based on past information using the estimation model and the isolated distribution based on the current observation. A smaller value of this metric indicates a significant discrepancy between the distribution inferred from past information using the estimation model and the distribution derived from current observations. To extend this metric for evaluating each detected non-local event, we define the weighted average of the 95% HPDROR between the one-step prediction distribution and the isolated distribution as shown in equation (**??**). As before, we include an interval in the evaluation set only if it contains at least one spike; otherwise, it is excluded from consideration.

The third metric we define is based on predictive checking. Specifically, as shown in (4), the predictive distribution $p\left(\vec{m}_k \mid \vec{m}_{0:(k-1)}, \Delta N_{0:k}^{(1:E)}\right)$ is calculated by the integral of one-step predictor distribution $p\left(x_k \mid \vec{m}_{0:(k-1)}, \Delta N_{0:(k-1)}^{(1:E)}\right)$ and the observation process $p\left(\vec{m}_k \mid x_k, \Delta N_k^{(1:E)}\right)$.

A low probability indicates that the current observation is unlikely under the distribution inferred from prior information using the assumed estimation model, suggesting potential inconsistencies in local model fitting. To extend this metric for evaluating each detected non-local event, we adopt the same formulation as in (**??**), where the predictive distribution is specified as $p\left(\vec{m}_k \mid \vec{m}_{0:(k-1)}, \Delta N_{0:k}^{(1:E)}\right)$. As before, an interval is included in the evaluation only if it contains at least one spike; otherwise, it is excluded from consideration.

3.3 Implementation Details

We begin with a simple toy example to develop intuition about each of the local goodness-of-fit metrics. We simulated a second movement using the generative model defined above. In this second, we assume that the rat is in the non-local state with small variance in the interval from 100 to 200 ms. From 300 to 400 ms, the rat is in the local state with some non-spatial hippocampal spikes. From 500 to 600 ms, the rat is in the non-local state with large variance, drift diffusion and local hippocampal spike rate. From 700 to 800 ms, the rat is in the non-local state with drift diffusion and high hippocampal spike rate. Then we simulated the movement trajectory o_k starting from the position $o_0 = 0$ cm using a random walk with standard deviation 0.3 cm for each time step. The range of the uniform distribution in (6) was from -10 to 10 cm. When $s_k = 1$ or 4, the random walk standard deviation is set $\sigma = 0.3$ cm. As for the larger random walk standard deviation when $s_k = 3$, we set it to $\sigma' = 1$ cm. When $s_k = 3$ or 4, the drift is set to $\beta = 150$ cm per second. The time step in our simulation was chosen to be 1 ms. The path was bounded between -10 cm and 10 cm by fixing any movements that would have extended beyond these bounds at the boundary. We sampled the semi-latent state $x_k|s_k$ using

(6). In order to match real trajectories, all the simulated spatial trajectories were smoothed.

We simulated the spike times and waveforms using a set of joint mark intensity functions as described in equation (7) that included two modes corresponding to two place cells for each tetrode. For the simple simulation scenario, we chose $\sigma_x = 1$ and $\sigma_m = 1$. When $s_k \neq 3$, we set $\alpha^i_{s_k,j} = 3.5$ for $i = 1, \cdots, 5$ and $j = 1, 2$. Otherwise, $\alpha^i_{s_k,j}$ was set to 1.5. When $s_k \neq 2$, we set $\omega^i_{s_k} = 0$ for $i = 1, \cdots, 5$. Otherwise, $\omega^i_{s_k}$ was set to 0.9. The other parameters are listed in Appendix.

The locality state was initially set in a local state, $\Pr(s_0 = 0) = 1$. By definition, when $s_0 = 0$, the distribution of x_0 is a delta function at the rat's linearized position. For the state transition probability for s_k, we used $Pr\left(s_k = 1 \mid s_{k-1} = 0\right) = 0.0001$ and $Pr\left(s_k = 1 \mid s_{k-1} = 1\right) = 0.95$.

The latent variable x_k is one-dimensional and s_k is discrete. Since there is no analytic solution of the cumulative distribution of latent state x_k, we employed numerical integration methods, specifically Riemann sum approximation to estimate the solution. The KL divergence was also calculated by the Riemann sum approximation. As for HPDROR, we first sorted each distribution we might use by the value of probability mass and then found the HPDR. Finally, we calculated the HPDROR through the HPDR we have defined. Finally, for predictive checking, since there was no analytic solution for $p\left(\vec{m}_k \mid \vec{m}_{0:(k-1)}, \Delta N^{(1:E)}_{0:k}\right)$, we used a two-stage monte carlo procedure to sample x_k under the one-step prediction distribution and $\vec{m}_k$ under the predictive distribution. By doing this, we estimated the likelihood of observing $\vec{m}_k$ under the predictive distribution.

In addition, based on the definition in (6), in each scenario, we simulated 120 non-local events with small variance and 120 local events with some non-spatial hippocampal spikes. We simulated the rat's movement for one minute, segmenting time into 100 ms intervals. Within each second, we designated the second and sixth intervals as the non-local state with small variance, i.e. $s_k = 1$, while the fourth and eighth intervals were set as the local state with non-spatial hippocampal spike data, i.e. $s_k = 2$. The remaining intervals were set as the local state, i.e. $s_k = 0$. In addition, we changed the random walk standard deviation parameter σ in (6) or the hippocampal spiking rate parameter $\alpha^i_{s_k,j}$ in (7) to generate the data under different scenarios and evaluated the two kinds of events. All the other parameters used to generate these events were the same as described previously. The parameters for the estimation model are adjusted in the value of σ or α^i_j. We further evaluated two distinct pairs of events: non-local events with small variance vs. non-local events with large variance, drift diffusion and low hippocampal spike rate and non-local events with small variance vs. non-local events with drift diffusion and high hippocampal spike rate. Following a similar method, we generated 120 events for each kind of event. To explore different scenarios, we changed σ and $\alpha^i_{s_k,j}$, while keeping all other parameters consistent with those used in the previous simulation study. We used the period-level metrics to evaluate the type I and type II error rates.

3.4 Simulation Results

Fig. 3 shows the decoding and evaluation results for the simulated data. Several detected non-local events under the misspecified estimation model are shown. The first row shows a heat plot of the filter probability distribution for the semi-latent continuous variable x_k. The red line shows the actual value of semi-latent variable x_k and the green line shows the observed

Figure pending: simulation-results plot.

Figure 3: Decoding and evaluation results for simulation study: the first row shows heat plot of the filter probability distribution for the semi-latent continuous variable, x_k. The red line is the actual trajectory of the semi-latent variable. The green line is the observed linearized position. The shaded area shows the marginal filter distribution of x_k. The second row shows the square root of the KL divergence between the filter distribution of x_k and the isolated distribution at each spike time. The third row shows the 95% HPDROR between the isolated distribution of x_k and one-step prediction distribution at each spike time. The fourth row shows the estimated likelihood under the predictive distribution of $\vec{m}_k$ for the observed spikes at each spiking moment. The red dotted line shows a significance level of 0.05. The blue boxes illustrate the content from each detected non-local event.

linearized position. When the animal is in the local state, x_k is the same as the observed linearized position. When the locality state changes from local to non-local, the representation of x_k jumps rapidly to a new location. When the model assumption is correct, the filter distribution should align with the true value of x_k, as we see in the first detected non-local event. As for the second detected non-local event, although the rat is in a local state, we still obtain numerous non-local estimates due to the incorrect model for the joint mark intensity function. As for the third detected non-local event, owing to our reliance on an overly simplistic estimation model and the scarcity of hippocampal spikes, noticeable discrepancies arise between our estimates and the true values. As for the last detected non-local event, this estimate is derived from a simpler estimation model. Given the smaller gap between the generative model and estimation model with the ample information provided by hippocampal spikes, our estimate closely aligns with the true value.

The second row shows the KL divergence between the filter distribution of x_k and isolated distribution at each spike time. When a non-local event is detected, it tends to decrease, apart from a few spikes at the beginning and end of the event—indicating state transitions. When the locality state is estimated as local, the filter distribution of x_k remains confined to the current observed linearized position. In contrast, when the state is estimated as non-local, the filter distribution extends beyond the current location, reaching other plausible positions. This broader spread allows for greater overlap with the spatial information implied by the current observation. For the first detected non-local event, since we used the same generative and estimation model, the KL divergence decreased during the event except the spikes at the beginning and end of the event. For period-level evaluation based on this metric, we cannot reject the assumption that this event is a non-local event with small variance. During the second detected event, there are also a substantial number of hippocampal spikes with large KL divergence values occurring. This suggests that our estimation model does not align with the dynamics represented by these spikes. Based on our period-level evaluation, we have enough evidence to reject the assumption of a non-local event with small variance. In the third detected non-local event, there are numerous moments with large KL divergence values. This indicates a discrepancy between these spikes and prior spiking information accumulated over time using our estimation model, suggesting that our model may not accurately capture the dynamics of observed spikes, making the decoding results unreliable. Based on our period-level evaluation, we reject the assumption that this event is a non-local event with small variance. In the last detected non-local event, the values remain small. Although our estimation model differs from the generative model, the abundance of hippocampal spikes leads to accurate estimates despite this misspecification. This metric indicates that our result remains relatively reliable. For the period-level evaluation, we cannot reject the assumption of a non-local event with small variance.

The third row shows the 95% HPDROR between the isolated distribution of x_k and one-step prediction distribution at each spike time. For the first detected non-local event, except for some spikes at the beginning and end of the event, the ratio is high, which means that the spikes we observed are consistent with the previous spiking information based on the estimation model, and we have not detected inconsistency here. Based on this metric, at the period level, we cannot reject that this is a non-local event with small variance. In the second detected non-local event, there are numerous moments where this ratio is extremely low, casting doubt on whether the observed hippocampal spikes align with our assumed model. Based on our period-level evaluation, we can reject that this is a non-local event with small variance. The third detected non-local event contains only a few spikes. Additionally, certain moments exhibit very low ratios, lowering the overall average ratio during this period. For the period-level evaluation, we reject that the event is a non-local event with small variance. The last detected non-local event closely resembles the first. Except for a few observed spikes at the beginning and end that exhibit lower ratios, the ratio remains high, leading us to believe that our estimation model aligns well with the generative model. Although this event is derived from a misspecified model, based on period-level metric, we cannot reject that it is a non-local event with small variance.

The last row shows the estimated likelihood under the predictive distribution of $\vec{m}_k$ based on one-step prediction distribution of x_k combined with the observation process $p\left(\vec{m}_k \mid x_k, \Delta N_k^{(1:E)}\right)$ for the observed spikes at each moment. Although this value differs in meaning from HPDROR, both exhibit similar trends. In the first detected non-local event, apart from a few spikes with extremely low likelihood at the beginning and end, the remaining spikes have marks that are likely based on the estimated state. Based on period-level metrics, we cannot reject the assumption of a non-local event with small variance. The second detected non-local event contains a large number of spikes with extreme marks given the estimated state, making it difficult to believe they align with previous spiking information based on our estimation model. At the same time, this event yields a very low value in the likelihood based on period-level evaluation, we can confidently reject it as a non-local event with small variance. The third detected non-local event contains relatively few spikes, and among them, a considerable number have marks with low likelihood. This diminishes our confidence that these spikes align with our estimation model. Consequently, based on period-level evaluation, we have enough confidence to reject the assumption that this event is a non-local one with small variance. Finally, for the last detected non-local event, apart from the spikes at the beginning and end of the detected event, the spikes occurring at other times have marks with high likelihood. This strengthens our confidence that our estimation model aligns well with these observed spikes. Although the generative model of the latent variables in this event differs from our estimation model, this metric suggests that our decoding results remain reasonable. Consequently, based on period-level evaluation, we cannot reject it as a non-local event with small variance.

The results above are mainly based on hippocampal spikes observed at each spike time. Next, we will present the findings using period-level metrics to evaluate the detected non-local events, determining whether we have enough evidence to reject the assumption that they are non-local events with a simple random walk movement model.

Fig. 4 shows the type I and II error rates in evaluating non-local events. As the random walk standard deviation σ increases, the error rate of all methods rises accordingly. This occurs because, even if the estimation model is incorrect, a larger standard deviation in the random walk diminishes the influence of past spiking information. In this case, the distribution of x_k at each moment relies more heavily on the spike data at that specific time. As a result, it becomes challenging to detect discrepancies between the distribution derived from the current spike and the distribution accumulated from previous spike information using the estimation model. In addition, the accuracy of our evaluation is also constrained by the hippocampal spike rate. When the spike count is high, the error rate remains low; however, with fewer spikes, the error rate increases. Among the three metrics, the one based on credible regions demonstrates the greatest robustness, yielding the lowest error rate in most scenarios. In contrast, the other

Figure pending: Type I/II error-rate plot.

Figure 4: Type I and II error rates of evaluation of non-local events. In the first column, we show the type I error rate vs. the random walk standard deviation σ defined in both (6) and (9). In the second column, we show the type I error rate vs. the rate parameter $\alpha^{i}_{s_k,j}$ in joint intensity mark function as mentioned in (7). The third and fourth column shows the corresponding type II error rate. The black line shows the results based on weighted average KL divergence between the filter distribution of x_k and isolated distribution. The red line shows the results based on weighted average HPDROR between the one-step prediction distribution of x_k and isolated distribution while the blue line shows the results based on prior predictive checking for $\vec{m}_k$. (a) - (d) show the type I and II error rates for evaluating non-local events with small variance and local events with non-spatial hippocampal spike data. (e) - (h) show the type I and II error rates for evaluating non-local events with small variance and non-local events with large variance, drift diffusion and low hippocampal spike rate. (i) - (l) show the type I and II error rates for evaluating non-local events with small variance and non-local events with drift diffusion and high hippocampal spike rate.

two metrics exhibit lower stability but possess unique advantages depending on the specific scenario. In addition, the metric based on predictive checking seems to have high dependence on the spike rate parameter in the intensity function. Lower spike rates always lead to inaccurate evaluation. In contrast, the method based on the credible region overlap ratio demonstrates greater robustness, consistently maintaining a lower type I error rate across various situations even when the level of model misspecification is very low. Of course, when the true generative model and our estimation model are close, the type II error from all methods becomes relatively large.

4 Real Hippocampal Data Analysis

We analyzed publicly available hippocampal data from a single male Long Evans rat trained to perform a memory-guided spatial alternation task on a W-shaped maze. The data, recorded in the laboratory of Dr. Loren Frank at UCSF is available online at `https://datadryad.org/stash/dataset/doi:10.7272/Q61N7ZC3`. We extracted spiking data from 12 tetrodes located in the CA1 region during a single 15 minute trial of spatial navigation.

For the clusterless spike, we used a kernel based intensity model fit over the spiking data to estimate intensity function from a separate training set of the form:

$$\lambda_i\left(t, \vec{m}\right) = \frac{\sum_{i=1}^{N} \mathcal{K}(x_t - x_i, \vec{m} - \vec{m}_i)}{\sum_{j=1}^{K} \mathcal{G}(x_t - x_j)} \tag{12}$$

where N is the total number of spikes in the training set, $\mathcal{K}(x_t - x_i, \vec{m} - \vec{m}_i)$ is a 5-dimensional Gaussian kernel function (1 dimension for space and 4 for each mark dimension), centered at each observed spike location and mark in the training set, and $\mathcal{G}(x_t - x_j)$ is a 1-dimensional

Gaussian kernel function centered at each location visited at the K discrete time points in the training set. We used a bandwidth of 6 cm for the linearized position and 24 μV for the spike amplitude dimensions.

For the estimation model, we adopt the same one as used in the simulation study. Specifically, we treat the locality state s_k as a binary variable, adhering to the definition given in (6). We assume $Pr\left(s_k = 1 \mid s_{k-1} = 0\right) = 0.0001$ and $Pr\left(s_k = 1 \mid s_{k-1} = 1\right) = 0.95$. Likewise, x_k is assumed to follow the distribution outlined in (9). The uniform distribution is set to the span of the environment between 0 and 187 cm. The standard deviation of random walk is set to 3 cm. The likelihood function governing the observed hippocampal spikes in each time interval $[t_{k-1}, t_k)$ also conforms to the definition provided in (10). We assumed that the locality state was initially to be in a local state, $\Pr(s_0 = 0) = 1$. In order to reduce the computational burden, for the prior predictive checking for the waveform features $\vec{m}_k$ at each spiking moment, we used a bootstrap method to get many samples of waveform features from our training set based on the sampled latent variable x_k.

We show three decoding examples in this section: the first one seems to be a real non-local events with small variance while the second one is suspicious. The third one presents decoding results with longer duration , revealing one detected non-local event, while the remaining time is interpreted as local state.

Fig. 5 shows several decoding examples for both local and non-local events detected in this dataset. Panels (a)-(c) shows the filter probability of being in a non-local state with small variance at each instant. When there are many hippocampal spiking moments with non-local representations, the filter probability switches from a value near 0 to a value near 1, and persists

Figure pending: real-data decoding-examples plot.

Figure 5: Three decoding examples for both local and non-local events in real dataset: (a)-(c) shows the filter probability of non-local state. (d)-(f) shows heat plot of filter probability distribution for semi-latent continuous variable x_k. The green line is the observed linearized position. (g)-(i) shows the KL divergence between filter distribution of x_k and isolated prediction distribution at each spiking moment. (j)-(l) shows the HPDROR between isolated distribution of x_k and one-step prediction distribution at each spiking moment. (m)-(o) shows the estimated probability under the predictive distribution of $\vec{m}_k$ for the observed spikes at each spiking moment. The red dot line shows a level of 0.05.

at that high value throughout the non-local event, after which it rapidly switches back. The filter probability does dip occasionally, when the spiking data become more ambiguous. Panels (d)-(f) show a heat plot of the filter probability distribution for the semi-latent continuous variable x_k. When the state changes from local to non-local, the representation jumps rapidly to a new location. While the estimated locality state is local, the filter distribution for x_k is fixed at the observed linearized position.

Fig. 5(g)-(i) show the KL divergence between the filter distribution of x_k and the isolated distribution at each spike time. In panel (g), when a non-local event occurred, the divergence tended to become smaller because the spatial information contained in each spike is close to that from filter estimate, leading to a reduction in divergence. The consistently lower values observed in this event suggest a strong alignment with the predefined state-space model. However, in panel (h), there are more spikes with higher value in this metric, demonstrating uncertainty about the decoding results. In panel (i), while the discrete state suggests that the neural representation is local, certain spikes exhibit unusually large values, suggesting a potential inconsistency between our assumed estimation model and the information from these spikes. The spatial content contained in these spikes deviates from the observed linearized position, yet does not gather sufficient non-local information to alter the estimate of the locality state from local to non-local. Consequently, these moments, though identified as local states, reveal underlying issues in local model fitting.

Fig. 5(j)-(l) show the HPDROR between isolated distribution of x_k and one-step prediction distribution at each moment. In panel (j), apart from the spike at the end, the overlapping ratio between the two distributions remains high throughout. The spikes with lower ratios serve as indicators of state transitions. Panel (k) shows the ratios for another decoding example with many lower ratios during this detected non-local event. These represent inconsistency in the model leading to uncertainty about the decoding results. In panel (l), while the discrete state suggests that the neural representation is local, there are still moments when this ratio drops to a low value, indicating a potential mismatch between our assumed model and the spiking information. The spatial content from these spikes deviates from the local state, yet it may not accumulate sufficient non-local information to meaningfully alter the locality state. As a result, these moments reveal a lack of fit.

Fig. 5(m)-(o) show the estimated probability under the predictive distribution of $\vec{m}_k$ based on one-step prediction distribution of x_k combined with the observation process $p\left(\vec{m}_k \mid x_k, \Delta N_k^{(1:E)}\right)$ for the observed spikes at each spiking moment. In panel (m), there is only one spike with relatively low probability at the end of the non-local event, while maintaining a higher probability at other times. On the other hand, in panel (n), the frequent occurrence of lower probabilities diminishes our confidence in the decoding results. In panel (o), multiple instances of low probability under the estimated local state suggest potential issues with our local model fit. In general, the first detected non-local event should not be dismissed as a non-local event with small variance, much like the so-called hippocampal replay event. However, with the second detected non-local event, we gain a certain degree of confidence in rejecting it as a non-local event with small variance. As for the estimated local state in the third example, the spatial information embedded in the spikes not only corresponds to the local state but also includes information in favor of non-local state, raising doubts about the reliability of our decoding results.

Fig. 6 shows how the three local goodness-of-fit metrics at each spike time evolve with

Figure pending: real-data metric-summary plot.

Figure 6: Summary of the three local goodness-of-fit metrics for decoding results at each spike time vs. estimated filter probability of the non-local state based on real hippocampal spike data. (a) KL divergence (b) HPDROR (c) p-value for predictive checking of observed mark $\vec{m}_k$.

the estimated filter probability of the non-local state. In panel (a), we show how the KL divergence between the filter distribution of x_k and the isolated distribution at each spiking moment changes with the estimated filter probability of the non-local state. Surprisingly, we find that the spikes with the largest divergence mostly occur during periods when the model estimates that the neural representation is local to the rat's position. This suggests that the model does not often generate non-local trajectories when the hippocampus is really encoding the rat's position. Instead it seems that there are numerous hippocampal spikes that contain non-local information during periods that the model assumes they should be local. This could occur, for example, if the hippocampus switches between local and non-local representations much more rapidly than the model assumes. Another possibility is that a subset of hippocampal neurons have non-local spiking properties while a larger majority of neurons do reflect the rat's current location. In panel (b), we observe that the smallest HPDROR values are mostly clustered at times when the discrete state suggests that spike should be local. This corroborates the finding based on the KL divergence, suggesting that the biggest source of model misspecification is related to the assumption that the hippocampus represents the rat's current location homogeneously and consistently in this state. In panel (c), we observe similar results for the p-values arising from predictive checking of the spike waveform marks. The lowest p-values cluster at times when the model estimates hippocampal coding to be local, suggesting that we observe spikes from neurons we would not expect to fire at these locations if all of the neurons were representing the rat's actual location as the model assumes.

Based on the above results, all three metrics consistently indicate that there exist multiple spikes that do not reflect the rat's location occurring at times when the model predicts that they should. In contrast, when the estimated state is non-local, the KL divergence-based metric predominantly produces small values, whereas the other two metrics exhibit few small values. These results likely stem from limitations in our model assumptions. To transform from local state to non-local state, the model requires continuous non-local spikes with consistent spatial content—a stringent condition that enhances the credibility of identified non-local events. When the estimated state is local, not all hippocampal spikes exhibit local spatial content. Due to insufficient accumulation of non-local information, the locality state remains local, even when certain spikes actually contain non-local spatial content. This misspecification leads to problems with the model's local fit. Crucially, the local goodness-of-fit metrics we established are able to detect and capture these discrepancies effectively.

Similar to the previous toy example, we present scatter plots among the three metrics at each moment using the real hippocampal data with the assumed two-state model.

In Fig. 7, because a relatively large bin was used to decode the latent variable x_k, HPDROR exhibits stratification to some extent. In (a), we show the scatter plot between $D_{\mathrm{KL}}(\mathrm{f}_{\text{accumulated}}(x) \| \mathrm{f}_{\text{isolated}}(x))$

Figure pending: real-data metric scatter plots.

Figure 7: Scatter plots of the three metrics from real hippocampal data: (a) shows scatter plot between $D_{\mathrm{KL}}(\mathrm{f}_{\text{accumulated}}(x) \| \mathrm{f}_{\text{isolated}}(x))$ and $\textbf{HPDROR}(\mathrm{f}_{\text{accumulated}}(x), \mathrm{f}_{\text{isolated}}(x))$ at each moment, (b) shows that between $D_{\mathrm{KL}}(\mathrm{f}_{\text{accumulated}}(x) \| \mathrm{f}_{\text{isolated}}(x))$ and $Pr(\log(\mathrm{f}_{\text{predictive}}(y)) \geq \log(\mathrm{f}_{\text{predictive}}(\widetilde{y})))$, and (c) shows that between $\textbf{HPDROR}(\mathrm{f}_{\text{accumulated}}(x), \mathrm{f}_{\text{isolated}}(x))$ and $Pr(\log(\mathrm{f}_{\text{predictive}}(y)) \geq \log(\mathrm{f}_{\text{predictive}}(\widetilde{y})))$.

and $\textbf{HPDROR}(\mathrm{f}_{\text{accumulated}}(x), \mathrm{f}_{\text{isolated}}(x))$. When HPDROR is small, the KL divergence tends to be relatively larger, and in these cases, the two metrics provide relatively consistent assessments of the overlap between $\mathrm{f}_{\text{accumulated}}(x)$ and $\mathrm{f}_{\text{isolated}}(x)$. However, as HPDROR approaches 1, the KL divergence spans a wider range of values, making it unreliable for accurately evaluating the degree of overlap. In (b), we show the scatter plot between $D_{\mathrm{KL}}(\mathrm{f}_{\text{accumulated}}(x) \| \mathrm{f}_{\text{isolated}}(x))$ and $Pr(\log(\mathrm{f}_{\text{predictive}}(y)) \geq \log(\mathrm{f}_{\text{predictive}}(\widetilde{y})))$. Compared to HPDROR, the p-values of the predictive check follow a similar trend, though they span a wider range, which makes stratification less evident. In (c), we show the scatter plot between $\textbf{HPDROR}(\mathrm{f}_{\text{accumulated}}(x), \mathrm{f}_{\text{isolated}}(x))$ and $Pr(\log(\mathrm{f}_{\text{predictive}}(y)) \geq \log(\mathrm{f}_{\text{predictive}}(\widetilde{y})))$. Consistent with the results of toy examples, the two metrics exhibit similar trends. Although the p-values generated by the predictive check vary over a wider range, they remain broadly comparable, suggesting that both metrics play a similar role in assessing the degree of overlap between $f_{\text{accumulated}}(x)$ and $f_{\text{isolated}}(x)$.

Drawing on both the previous toy example and the analysis of real hippocampal data, we compared three metrics. Because KL divergence provides only an indirect measure for overlap between $f_{\text{accumulated}}(x)$ and $f_{\text{isolated}}(x)$, its values may not reliably capture the true extent of overlap between the two distributions. In contrast, both HPDROR and the p-values of the predictive check capture the degree of overlap between $f_{\text{accumulated}}(x)$ and $f_{\text{isolated}}(x)$ with considerable accuracy. More importantly, the two metrics yield largely consistent assessments of this overlap. Given the higher computational cost of the predictive check, we therefore favor HPDROR as a faster and equally reliable measure to evaluate the degree of overlap.

5 Discussion & Conclusion

State space modeling has become a powerful tool for describing the latent dynamics of neural systems. However, validating these models is inherently difficult because latent states are unobservable and many models are purposefully simplified. To address this, we developed multiple local goodness-of-fit metrics that compare the instantaneous information received from each spike to the aggregated information computed based on recent spiking activity and the state-space model assumptions. Rather than dismissing a model in its entirety, these metrics are able to assess the consistency between immediate observations and the information accumulated through the state-space structure, providing a more granular assessment of decoding reliability.

We defined three classes of metrics: F-divergence measures (e.g., KL divergence), overlap ratios between credible regions (HPD overlap), and predictive checks. While KL divergence is computationally simple, it is sensitive to distribution tails and lacks a clear upper bound for assessing similarity. In contrast, the HPD overlap offers a more robust geometric measure of similarity, consistently maintaining lower error rates across various conditions. Predictive checks provide a direct statistical assessment through observation-level p-values, but are more computationally intensive. While each metric has particular advantages, when used in combination, they provide a more complete perspective on decoding reliability.

We illustrated the application of these metrics to simulated and real spike sorted data from rat hippocampus. These same metrics naturally extend to other neural data types when incorporated into state space models. In particular, recent work has shown that clusterless neural spiking models can substantially reduce bias and variance in neural decoding and estimation problems when compared to spike sorted models. The metrics, as defined in Section 2, naturally extend to clusterless models. In that case, a lack of fit for any particular spike suggests that a spike with that waveform was unexpected based on the accumulated information from the state space model up to that point in time.

For simplicity, in this paper, we focused on comparing the instantaneous information in the likelihood of each spike to the aggregated information from the past contained in the one-step prediction density. A natural, powerful extension of these methods is obtained by replacing the one-step prediction density with any posterior density derived from the state-space framework. In particular, replacing the one-step prediction density with a smoothing density (e.g. from a fixed-interval smoother over the observation period) would allow us to leverage both past and future data to increase statistical power in identifying model misspecification.

One potential challenge for applying these methods broadly has to do with the computational burden of computing these metrics as the dimensionality of the state process increases. In all of our examples, we computed the KL divergence measure and the HPD overlap by partitioning the state space into a lattice and computing the likelihoods and prediction densities at each point in that lattice. As the dimension of the state increases, the size of that lattice will grow exponentially. While the predictive checks are the most computationally burdensome in lower state dimensions, the fact that we are sampling from the state-space rather than computing values over a lattice means that in higher dimensions the predictive checks will become more computationally efficient. It also suggests that we may be able to improve the efficiency of the KL and HPD overlap metrics by using sampling-based estimation methods. This will not solve the curse of dimensionality, as the number of samples required for accurate estimates will likely still grow with dimension, but it is likely to allow for efficient estimation in moderate dimensions. Another approach for dealing with this issue may be to compute these consistency metrics in lower-dimensional subspaces of the state space. How to identify subspaces that retain the statistical power of the original space is a topic for future exploration.

Another potential challenge relates to identifying time-scales for which to assess for lack of fit. These methods allow us to focus in on time scales as short as individual spikes, and for our place cell decoding examples individual spikes are sufficiently informative that we are often able to detect local model misfit at this time-scale. For other neural systems, it may be the case that local model errors are more likely to be detected over longer timescales. In this paper we discuss how to extend these goodness-of-fit measures to longer time intervals, but determining what time scales are most appropriate may require relying about prior knowledge about the system, or exploring the values of these metrics across multiple time scales.

We believe that these local goodness-of-fit tools provide an important step toward making inferences from state-space neural models more reliable and interpretable. By pinpointing specific moments when the model assumptions break down, researchers can be more confident in their inferences. Ultimately, these measures empower us to refine our understanding of neural dynamics and deepen our understanding of how the brain encodes and processes information.

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